Write your name here

Surname	Other names
Master Functional Skills	Model Solutions (MFS)

Pearson Edexcel Functional Skills

Centre Number	Candidate Number
MFS-UK	FSM01/01

Mathematics

Level 1



11 – 15 June 2018
Time: 1 hour 30 minutes

Paper Reference
FSM01/01

You must have:
Pen, calculator, HB pencil, eraser, ruler graduated in cm and mm, protractor, compasses.

Total Marks
48 /48

My signature confirms that I will not discuss the content of the test with anyone until the end of the 5 day test window.

Signature: Master Functional Skills (MFS)

Instructions

- Use a **black** ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Sign the declaration.
- Answer **all** questions.
- Answer the questions in the spaces provided
  – there may be more space than you need.
- **Calculators may be used.**

Information

- The total mark for this paper is 48.
- The marks for each question are shown in brackets
  – use this as a guide to how much time to spend on each question.
- **You must show clearly how you get your answers because marks will be awarded for your working out.**
- **Check your working and your answers at each stage.**
- **This sign shows where marks will be awarded for showing your check.**

Advice

- Read each question carefully before you start to answer it.
- Keep an eye on the time.

Turn over ▶

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SECTION A: New car

Answer all questions in this section.

Write your answers in the spaces provided.

1 Michael wants to buy a new car.

He needs to pay

- a cash deposit of £5875
- £229.20 each month for 24 months.

(a) Work out the total amount Michael will pay.
Show a check of your working. **(4)**

Use the box below to show clearly how you get your answer.

Step 1: Calculate total monthly payments for 24 months:

- *24 months × £229.20 = £5,500.80 [M1]*

Working: 20 × £229.20 = £4584; 4 × £229.20 = £916.80 -> 4584 + 916.80 = £5,500.80

Step 2: Add the cash deposit to the monthly total:

- *Total = £5,500.80 + £5,875 = £11,375.80 [M1]*

Step 3: State final answer in correct money notation:

- *Total amount Michael will pay = £11,375.80 [A1]*

Write your check in the box below.

☑

Reverse Calculation & Estimation Check:

- *Reverse subtraction: £11,375.80 − £5,875 = £5,500.80* ✓
- *Reverse division: £5,500.80 ÷ 24 = £229.20 (matches monthly payment)*
- *Estimation check: £230 × 24 = £5,520; £5,520 + £5,875 = £11,395 ≈ £11,375.80*

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Michael needs to buy car insurance.
He finds this information.

insurance company	cost each month	excess	legal expenses	breakdown cover
A	£84.99	£350	yes	no
B	£88.90	£300	yes	yes
C	£91.50	£250	no	no
D	£94.33	£250	no	yes
E	£95.99	£250	yes	yes
F	£102.50	£300	yes	yes

Michael wants car insurance that

- has breakdown cover
- has an excess of less than £300
- costs the lowest amount each month.

(b) Which insurance company should Michael use?

(1)

Use the box below to show clearly your answer.

Filter options using Michael's 3 requirements:

1. Breakdown cover required: rules out Companies A and C (No breakdown).

2. Excess less than £300: rules out Companies B and F (excess = £300, not < £300).

3. Lowest monthly cost between D (£94.33) and E (£95.99): £94.33 < £95.99.

Michael should use Company D [B1]

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Michael wants to buy winter tyres for the car.

The normal price for the tyres is £400
There is a discount of 15% off the normal price.

Michael thinks 15% of £400 is £50

(c) Is 15% of £400 equal to £50?

(2)

Use the box below to show clearly how you get your answer.

Step 1: Calculate 15% of £400 normal price:

- *10% of £400 = £400 ÷ 10 = £40*
- *5% of £400 = £40 ÷ 2 = £20*
- *15% of £400 = £40 + £20 = £60 [M1]*

Alternative method: 0.15 × £400 = £60 (or 15/100 × 400 = £60)

Step 2: Compare with Michael's estimate of £50:

- *£60 ≠ £50 (it is £10 more) -> No, Michael is NOT correct. [A1]*

(Total for Question 1 is 7 marks)

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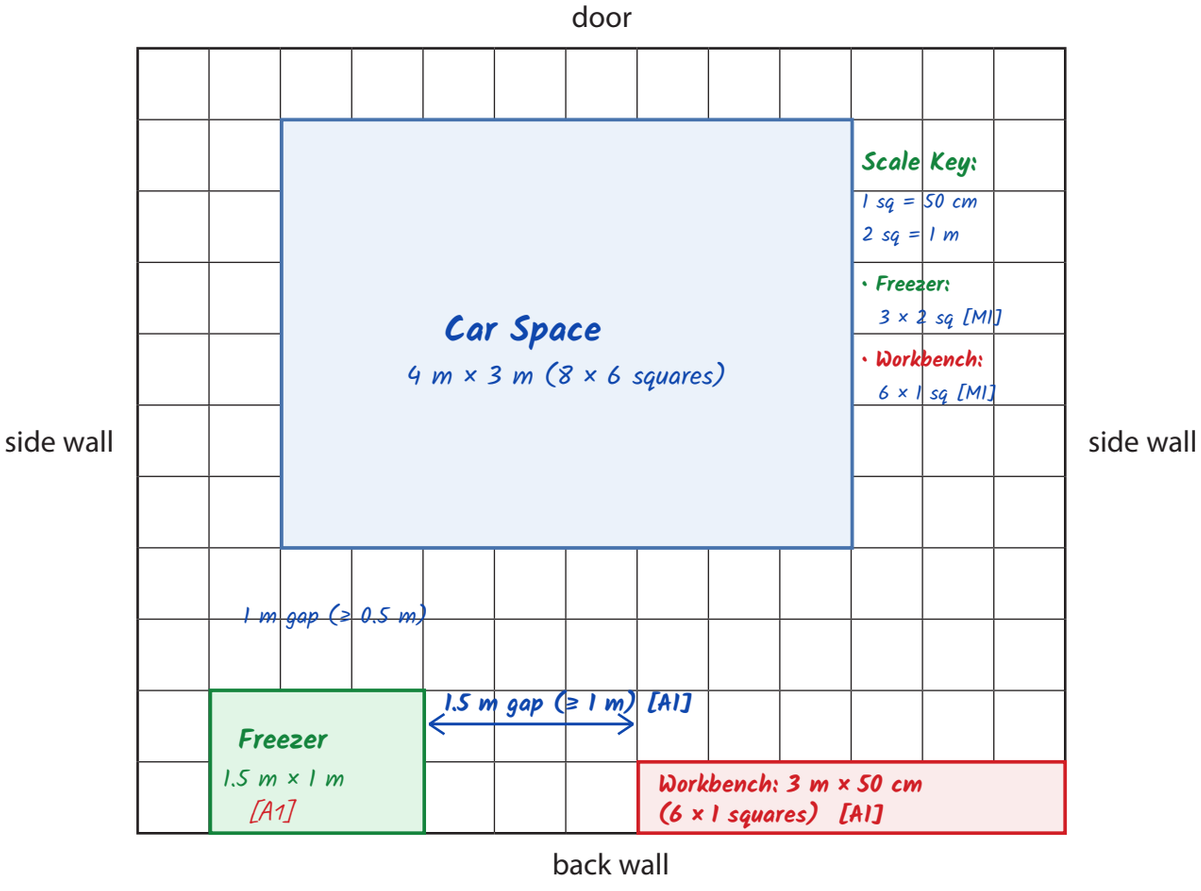
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2 Michael parks the car in his garage.

He wants to know if there is also space for a freezer and a workbench in the garage.

Michael makes a scale drawing of the garage floor.

Scale drawing of the garage floor



Key 1 square on the grid represents 50 cm by 50 cm on the garage floor

The car takes a rectangular space 4 m by 3 m.

The freezer needs a rectangular space 1.5 m by 1 m.
The workbench needs a rectangular space 3 m by 50 cm.

The longest side of the freezer and the longest side of the workbench must go along a wall.

There must be a gap of at least

- 50 cm between the space for the car and **any** other object
- 1 m between the space for the freezer and the space for the workbench.

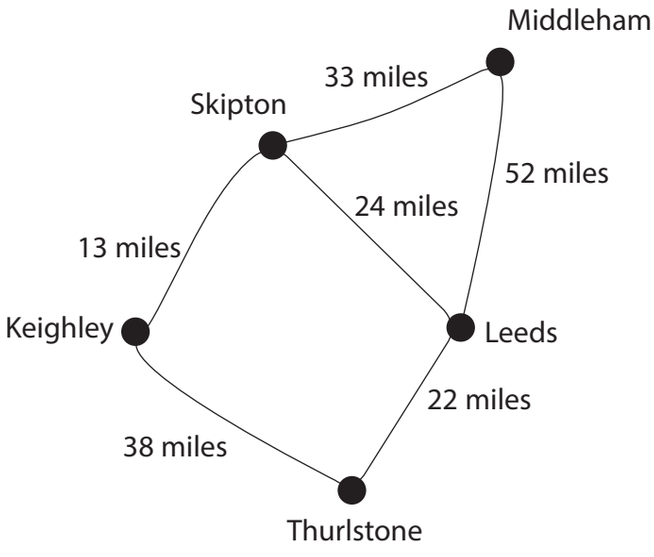
Draw the space for the freezer and the space for the workbench on the grid above. **(4)**

(Total for Question 2 is 4 marks)

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3 Michael plans a trip in his new car.
He wants to visit Skipton and Middleham.

Michael will start and end his trip in Thurlstone.
He draws this sketch to show distances between places.



Michael wants to use the sketch to find the shortest route.
He knows the cost of petrol to drive 10 miles is £1

Michael thinks the cost of petrol to drive the shortest route will be less than £16

Is Michael correct?
Show why you think this. **(5)**

Use the box below to show clearly how you get your answer.

Step 1: Identify routes from Thurlstone (T) visiting S & M and returning:

- *Route 1: T -> Leeds -> Skipton -> Middleham -> Leeds -> T*
  Distance = 22 + 24 + 33 + 52 + 22 = 153 miles [M1] (Shortest)
- *Route 2: T -> Keighley -> Skipton -> Middleham -> Leeds -> T*
  Distance = 38 + 13 + 33 + 52 + 22 = 158 miles
- *Route 3: T -> Keighley -> Skipton -> Middleham -> Skipton -> Keighley -> T*
  Distance = 38 + 13 + 33 + 33 + 13 + 38 = 168 miles

Step 2: Calculate petrol cost for shortest route (153 miles @ 10 miles/£1):

- ***Petrol cost = 153 ÷ 10 = £15.30 [M1]***
- *Alternative: £16 buys 16 × 10 = 160 miles (160 miles > 153 miles) [M1]*

Step 3: Conclusion:

- ***£15.30 < £16 -> Yes, Michael is correct. [A1]***

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Comprehensive Route Comparison Table & Distance Matrix:

Route	Full Path Description	Distance	Petrol Cost
Route 1 (Best)	T -> Leeds -> Skipton -> Mid -> Leeds -> T	153 miles	£15.30
Route 2	T -> Keighley -> Skipton -> Mid -> Leeds -> T	158 miles	£15.80
Route 3	T -> Leeds -> Skipton -> Mid -> Skip -> Leeds -> T	158 miles	£15.80
Route 4	T -> Keighley -> Skipton -> Mid -> Skip -> K -> T	168 miles	£16.80

Why Route 1 is the most fuel-efficient route:

1. Going from Thurlstone to Skipton via Leeds takes 22 + 24 = 46 miles, whereas going via Keighley takes 38 + 13 = 51 miles (saves 5 miles!).
2. Returning from Middleham to Leeds direct takes 52 miles, whereas backtracking through Skipton takes 33 + 24 = 57 miles (saves 5 miles!).

Petrol Budget Breakdown:

- **Maximum miles with £16 budget = £16 × 10 miles/£1 = 160 miles.**
- **Miles remaining in tank = 160 − 153 = 7 miles of spare fuel.**
- **Change from £16 = £16.00 − £15.30 = £0.70 (70p saved).**

(Total for Question 3 is 5 marks)

SECTION B: The exhibition

Answer all questions in this section.

Write your answers in the spaces provided.

4 Kay is organising an exhibition on health and beauty.
The exhibition will be open from 11 am to 4 pm.

There will be

3 different exhibition rooms with presentations
a lunch break for 45 minutes
a coffee break for 30 minutes.

Lunch and coffee will be served in the restaurant.

Kay wants people to spend an equal amount of time in each exhibition room.
She thinks this means people will spend at least 1 hour 20 minutes in each exhibition room.

(a) Is Kay correct?
Show why you think this. **(4)**

Use the box below to show clearly how you get your answer.

Step 1: Calculate total opening time between 11 am and 4 pm:

- *11:00 am to 4:00 pm = 5 hours = 5 × 60 = 300 minutes [M1]*

Step 2: Calculate total break time for lunch and coffee:

- *Lunch break = 45 mins; Coffee break = 30 mins*
- *Total break time = 45 + 30 = 75 minutes (1 hour 15 minutes) [M1]*

Step 3: Calculate available time for the 3 exhibition rooms:

- *Available presentation time = 300 mins − 75 mins = 225 minutes*
- *Time per room = 225 ÷ 3 = 75 mins = 1 hr 15 mins [M1]*

Step 4: Compare with Kay's target of at least 1 hour 20 minutes (80 minutes):

- *1 hour 15 mins (75 mins) < 1 hour 20 mins (80 mins).*
- *People only get 1 hour 15 minutes in each room (5 minutes less than 1 hr 20 min).*

Therefore, Kay is NOT correct (No). [A1]

Alternative check: 3 × 80 mins + 75 mins break = 315 mins (5h 15m)
11:00 am + 5h 15m = 4:15 pm, which exceeds 4:00 pm closing time.

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Kay wants to plan the displays in a room.

Each display needs a space of 3 m^2

Kay has a sketch of the room.
All corners are right angles.

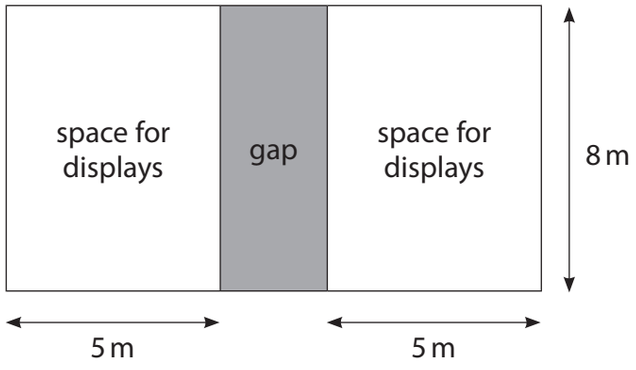


diagram **not** accurately drawn

Kay thinks there is enough space for at least 25 displays.

(b) Is there enough space for at least 25 displays in the room?

(4)

Use the box below to show clearly how you get your answer.

Step 1: Calculate total area available for displays in the room:

- *Left display rectangle = 5 m × 8 m = 40 m^2 [M1]*
- *Right display rectangle = 5 m × 8 m = 40 m^2*
- *Total display area = 40 m^2 + 40 m^2 = 80 m^2 [M1]*

Step 2: Calculate total area needed for 25 displays:

- *Each display needs 3 m^2:*
- *Area needed = 25 displays × 3 m^2 = 75 m^2 [M1]*

Step 3: Compare available area with required area:

- *80 m^2 available > 75 m^2 needed (5 m^2 surplus display space).*
- *Maximum displays that could fit = 80 ÷ 3 = 26.6... -> 26 full displays.*

Therefore, Yes, there is enough space for at least 25 displays. [A1]

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Jermaine will have a perfume display in one of the rooms.
He has 2 litres of perfume to give away to visitors.

Jermaine has 120 empty bottles.
Each bottle can hold 15 ml of perfume when full.

(c) Does Jermaine have enough perfume to fill up all the bottles?
Show a check of your working. **(4)**

Use the box below to show clearly how you get your answer.

Step 1: Convert perfume volume to millilitres (consistent units):

- 1 litre = 1000 ml
- **2 litres = 2 × 1000 = 2000 ml [M1]**

Step 2: Calculate total perfume needed to fill 120 bottles:

- **120 bottles × 15 ml each = 1800 ml (or 1.8 litres) [M1]**

  Working: 100 × 15 = 1500; 20 × 15 = 300 -> 1500 + 300 = 1800 ml

Step 3: Compare available volume with volume needed:

- **1800 ml needed < 2000 ml available (200 ml of perfume leftover).**
- Alternatively: 2000 ml ÷ 15 ml = 133.3... bottles (enough for 133 bottles).

Therefore, Yes, Jermaine has enough perfume to fill all 120 bottles. [A1]

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Write your check in the box below.

☑

Reverse Calculation Check:

- **Reverse division: 1800 ÷ 15 = 120 bottles [B1]**
- Reverse division by bottles: 1800 ml ÷ 120 = 15 ml per bottle

(Total for Question 4 is 12 marks)

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5 Kay asks 84 people to complete a survey about the exhibition.
Only 21 people complete the survey.

Kay wants to know what 21 is as a fraction of 84

(a) Write down 21 as a fraction of 84
Write your answer as a fraction in its simplest form. **(2)**

Use the box below to show clearly how you get your answer.

Step 1: Write initial fraction of people who completed survey:

- Fraction = 21 / 84 [M1]

Step 2: Simplify fraction to simplest form:

- Divide numerator and denominator by 7: 21 ÷ 7 = 3, 84 ÷ 7 = 12 -> 3 / 12
- Divide numerator and denominator by 3: 3 ÷ 3 = 1, 12 ÷ 3 = 4 -> 1 / 4

Alternative direct simplification by HCF (21):

- 21 ÷ 21 = 1, 84 ÷ 21 = 4 -> 1 / 4

Final Answer = 1 / 4 [A1]

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Kay wants to record the ratings from the survey.
She starts to fill in the tally chart below.

rating	tally	number of people
poor	𝍸	2
ok	𝍸	4
good	𝍸 𝍸	6 [M1]
excellent	~~𝍸~~ 𝍸	9 [A1]

Kay still needs to put these ratings into the tally chart and complete the last column of the table.

poor	good	good	excellent	poor
excellent	excellent	good	good	excellent

(b) Complete the tally chart fully for Kay.

(2)

(Total for Question 5 is 4 marks)

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SECTION C: Snooker

Answer all questions in this section.

Write your answers in the spaces provided.

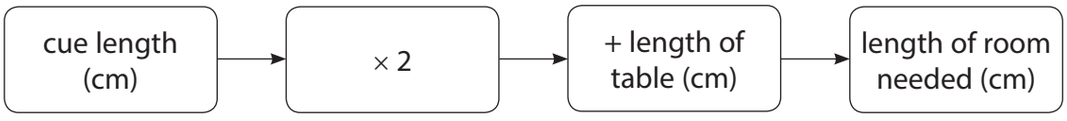
6 Talvin wants to put a snooker table in a room in his house.



The length of

- the snooker table is 270 cm
- the room is 5.5 m
- the cue Talvin uses to play snooker is 145 cm.

Talvin uses this rule to find out if the length of the room is enough to play snooker on this table.



(a) Is the length of this room enough to play snooker on this table?

(4)

Use the box below to show clearly how you get your answer.

Step 1: Convert room length to centimetres (consistent units):

- *Room length = 5.5 m = 5.5 × 100 = 550 cm [M1]*

Step 2: Apply the rule to calculate minimum room length needed:

- *Cue length × 2 = 145 cm × 2 = 290 cm [M1]*
- *Add table length: 290 + 270 = 560 cm (5.6 m) [M1]*

Step 3: Compare room length with room length needed:

- *550 cm available < 560 cm needed (the room is 10 cm too short).*

Therefore, No, the length of this room is NOT enough. [A1]

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Talvin needs to buy some snooker chalk to use with the cue.
He compares these two offers.

Offer 1	**Offer 2**
4 pieces of chalk	6 pieces of chalk
£1.80	£2.28

Talvin thinks that each piece of chalk is 10p cheaper with offer 2

(b) Is each piece of chalk 10p cheaper with offer 2?
Show a check of your working. **(5)**

Use the box below to show clearly how you get your answer.

Step 1: Calculate unit price for Offer 1 (4 pieces for £1.80):

- *£1.80 ÷ 4 = £0.45 = 45p per piece of chalk [M1]*

Step 2: Calculate unit price for Offer 2 (6 pieces for £2.28):

- *£2.28 ÷ 6 = £0.38 = 38p per piece of chalk [M1]*

Step 3: Calculate difference in price per piece:

- *45p − 38p = 7p cheaper with Offer 2 [M1]*

Step 4: Compare with Talvin's claim of 10p cheaper:

- *7p ≠ 10p (Offer 2 is only 7p cheaper per piece, not 10p).*

Therefore, Talvin is NOT correct (No). [A1]

Write your check in the box below.

☑

Reverse Multiplication Check:

- *Reverse Offer 2: 6 × 38p = £2.28 [B1]*
- *Reverse Offer 1: 4 × 45p = 180p = £1.80 (both match total offer costs)*

✓

(Total for Question 6 is 9 marks)

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7 Talvin is having a snooker competition with his friends.
There are 5 people in the competition

Paul, Ravi, Sam, Talvin and Violet.

Every person must play one game with each of the other people.

There is only one snooker table.
No one can play more than two games in a row.

Talvin plans who plays who in each game in the competition.
Paul and Ravi will play game one.

(a) Plan who plays who in the rest of the games in the competition.

(3)

Use the box below to show your answer clearly.

Game	Players
1	Paul and Ravi
2	Sam and Talvin
3	Violet and Paul
4	Ravi and Sam
5	Talvin and Violet
6	Paul and Sam
7	Ravi and Talvin
8	Sam and Violet
9	Paul and Talvin
10	Ravi and Violet

- All 10 pairings scheduled (every player plays 4 games) [M1]
- Constraint verified: No player plays more than 2 games in a row [A1]

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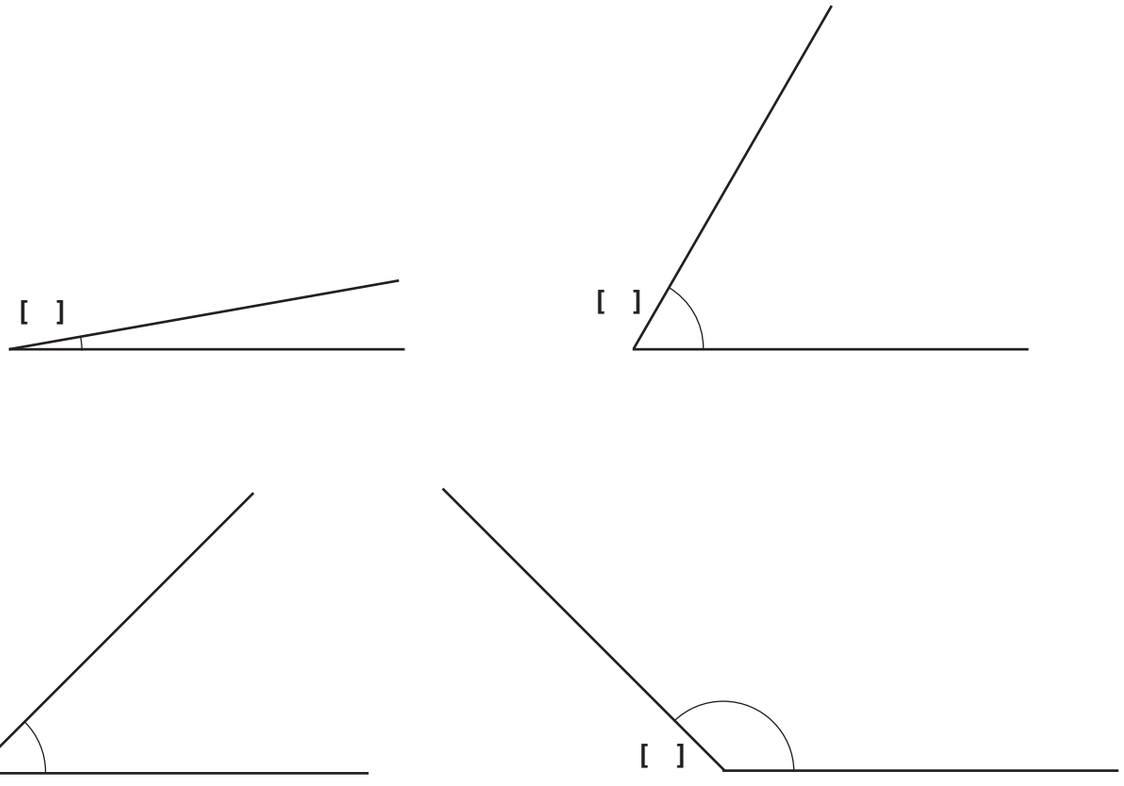
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Talvin wants the white ball to move at a 45° angle to the edge of the table.

(b) Which of these angles is 45°?

(1)

Tick [✓] to show your answer.



45° (half of a 90° right angle) [B1]

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The friends play 4 games of snooker in $2\frac{1}{2}$ hours.

The first game lasts for 45 minutes.

Talvin says

"Each of these games lasted a mean time of 45 minutes."

(c) Is Talvin correct?
Show why you think this. **(3)**

Use the box below to show clearly how you get your answer.

Step 1: Convert total time into minutes (consistent units):

- **2 1/2 hours = (2 × 60) + 30 = 150 minutes [M1]**

Step 2: Calculate the mean time for all 4 games:

- *Mean time = Total minutes ÷ Number of games*
- **Mean time = 150 ÷ 4 = 37.5 minutes per game [M1]**

Step 3: Compare with Talvin's statement of 45 minutes:

- **37.5 minutes < 45 minutes (the actual mean was 7.5 minutes shorter).**
- *Alternative: 4 games × 45 minutes = 180 minutes (3 hours) > 150 minutes.*

Step 4: Conclusion:

Therefore, Talvin is NOT correct (No). [A1]

(Total for Question 7 is 7 marks)

TOTAL FOR PAPER IS 48 MARKS

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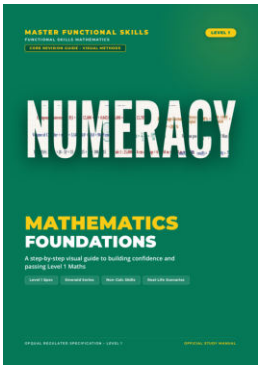
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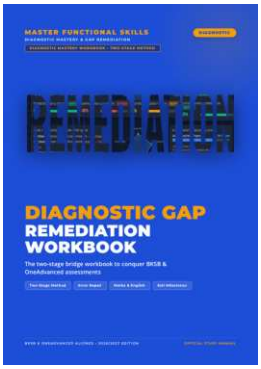


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